

FTML Exercices 5 solutions

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TABLE DES MATIÈRES

1 Logistic regression

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1 LOGISTIC REGRESSION

1]

$$\begin{aligned}\forall z \in \mathbb{R}, \sigma'(z) &= \left(-\frac{1}{(1+e^{-z})^2} \right) (-e^{-z}) \\ &= \frac{1}{1+e^{-z}} \frac{e^{-z}}{1+e^{-z}} \\ &= \frac{1}{1+e^{-z}} \frac{e^{-z}e^z}{(1+e^{-z})e^z} \\ &= \frac{1}{1+e^{-z}} \frac{1}{1+e^z} \\ &= \sigma(z)\sigma(-z)\end{aligned}\tag{1}$$

2] We compute the second order derivative.

$$\begin{aligned}\frac{\partial l}{\partial \hat{y}}(\hat{y}, y) &= \frac{-ye^{-\hat{y}y}}{1+e^{-\hat{y}y}} \\ &= \frac{-ye^{-\hat{y}y}}{1+e^{-\hat{y}y}} \frac{e^{\hat{y}y}}{e^{\hat{y}y}} \\ &= \frac{-y}{e^{\hat{y}y} + 1} \\ &= -y\sigma(-\hat{y}y)\end{aligned}\tag{2}$$

Hence,

$$\begin{aligned}\frac{\partial^2 l}{\partial \hat{y}^2}(\hat{y}, y) &= -y\sigma(-\hat{y}y)\sigma(\hat{y}y) \times -y \\ &= y^2\sigma(-\hat{y}y)\sigma(\hat{y}y) > 0\end{aligned}\tag{3}$$

Hence, the second-order derivative is strictly positive, and $l(\hat{y}, y)$ is strictly convex in its first argument.

3] We introduce the following functions :

$$\begin{aligned}g_i &= \begin{cases} \mathbb{R}^d \rightarrow \mathbb{R} \\ \theta \mapsto l(x_i^T \theta, y_i) \end{cases} \\ u_i &= \begin{cases} \mathbb{R} \rightarrow \mathbb{R} \\ \hat{y} \mapsto l(\hat{y}, y_i) \end{cases}\end{aligned}$$

$$v_i = \begin{cases} \mathbb{R}^d \rightarrow \mathbb{R} \\ \theta \mapsto x_i^T \theta \end{cases}$$

Then, $\forall i$

$$l(x_i^T \theta, y_i) = g_i(\theta) = (u_i \circ v_i)(\theta) \quad (4)$$

a] (convexity) It is sufficient to show that each $g_i : \theta \rightarrow l(x_i^T \theta, y_i)$ is convex, because the sum of convex functions is convex. By definition, (equation 4), g_i is a convex function u_i applied to a linear mapping v_i , which proves that g_i is convex.

b] (gradient) By composition of the jacobian matrices,

$$L_{\theta}^{g_i} = L_{v_i(\theta)}^{u_i} L_{\theta}^{v_i} = u_i'(v_i(\theta)) L_{\theta}^{v_i} \quad (5)$$

Or equivalently :

$$\nabla_{\theta} g_i(\theta) = u_i'(v_i(\theta)) \nabla_{\theta} v_i(\theta) \quad (6)$$

We already know that $\nabla_{\theta} v_i(\theta) = x_i$.

In question 2, we have seen that $\forall y, \hat{y}$,

$$\frac{\partial l}{\partial \hat{y}}(\hat{y}, y) = -y \sigma(-\hat{y} y) \quad (7)$$

Hence,

$$\begin{aligned} u_i'(v_i(\theta)) &= -y_i \sigma(-v_i(\theta) y_i) \\ &= -y_i \sigma(-x_i^T \theta y_i) \end{aligned} \quad (8)$$

Finally,

$$\nabla_{\theta} g_i(\theta) = -y_i \sigma(-x_i^T \theta y_i) x_i \quad (9)$$

and

$$\begin{aligned} \nabla_{\theta} R_n(\theta) &= \frac{1}{n} \sum_{i=1}^n \nabla_{\theta} g_i(\theta) \\ &= \frac{1}{n} \sum_{i=1}^n -y_i \sigma(-x_i^T \theta y_i) x_i \end{aligned} \quad (10)$$